Arquiteturas de Alto Desempenho 2026/2027

Practical work guide AAD_P8 (2026-11-30)

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1 Objectives

Simulate some simple sequential circuits described using VHDL.

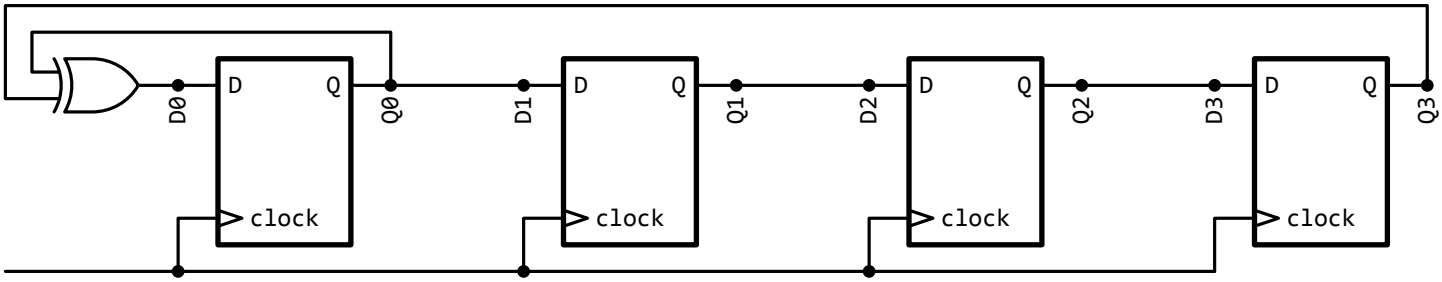
2 Exercise 1: D-type flip-flop

The D-type flip-flop is one of the most useful digital circuits. It can be used to store one bit of information. The file `d_flip_flop.vhd` contains a VLDH description of a simple D-type flip-flop. Study it. The file `ex1_tb.vhd` contains a test bench that can be used to generate waveforms that ilustrate how the flip-flop works. Load both to `edaplayground.com` and run the simulation.[1] As as alternative, if you have `ghdl` just do

```
$ make ex1.vcd
$ gtkwave ex1.vcd
```

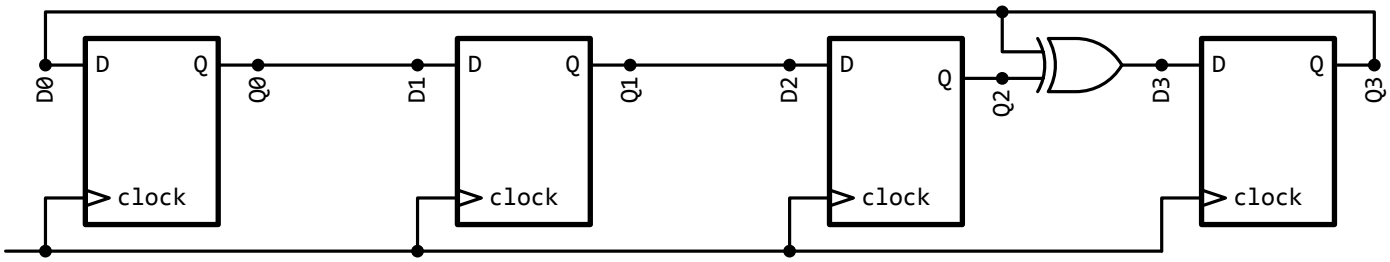
3 Exercise 2: more complex circuits using flip-flops

Create a VHDL entity that implements the following digital circuit.



Starting with all flip-flops in the `'1'` state, simulate the circuit for 50 clock cycles. The Q3 signal is periodic (actually, all signals are periodic). What is its period? Considering four consecutive clock cycles, what patterns in the Q3 signal are present?

Repeat the above but for this other circuit.



[1] As described in the previous practical work guide, to use EDA playground, https://edaplayground.com/, you will need to register first. After that you will have access to commercial software (free of charge when used for pedagogical purposes). It is, however, more convenient ans faster to use `ghdl` and `gtkwave`.

4 Exercise 3: dual-port memory

The file `dual_port_memory.vhd` contains the description of a dual port memory. It provides four implementations (arquitectures): either the reads are synchrounous or they are asynchronous, and either the reads read old data when the read address is equal to the write address or they read the new data. Study it with extra care. File `ex3_tb.vhd` contains a test bench that can be used to generate waveforms that ilustrate how the dual port memory works. Load both to `edaplayground.com` and run the simulation. As as alternative, if you have `ghdl` just do

```
$ make ex3.vcd
$ gtkwave ex3.vcd
```

In the second practical assignment you will need to adapt the dual port entity to your needs, so study it with extra care.

Advice: in order to make the simulation easier to understand, use first very small delays. In particular, edit the file `dual_port_memory.vhd` in order to make the delays a little smaller. For example, replace "`200 ps`" by "`10 ps`" (or less).